Creating the tables:

Employee table:

```
MariaDB [project_schema]>create table Employee (empNo int primary key, fName
varchar (20), lName varchar (20), address varchar (50), DOB date, sex ENUM('M'
,'F'), position ENUM ('Manager', 'Team Leader', 'Analyst', 'Software
Developer'),deptNo int);
Query OK, 0 rows affected (0.078 sec)

MariaDB [project_schema]> select * from Employee;
+-------+---------+-------+------------+------------+------+--------------------+-
-------+
| empNo | fName   | lName | address    | DOB        | sex  | position           |
deptNo |
+-------+---------+-------+------------+------------+------+--------------------+-
-------+
|     1 | John    | Smith | London     | 1990-01-10 | M    | Manager            |
    1 |
|     2 | Emma    | Brown | Manchester | 1992-03-15 | F    | Team Leader        |
    1 |
|     3 | Michael | Lee   | Birmingham | 1995-06-20 | M    | Analyst            |
    2 |
|     4 | Sarah   | White | Leeds      | 1993-09-12 | F    | Software Developer |
    2 |
|     5 | David   | Green | Liverpool  | 1991-11-05 | M    | Analyst            |
    3 |
|     6 | Anna    | Taylor | London    | 1994-02-18 | F    | Software Developer |
    3 |
|     7 | James   | Wilson | Oxford    | 1989-07-22 | M    | Manager            |
    4 |
|     8 | Sophia  | Moore | Cambridge  | 1996-12-30 | F    | Team Leader        |
    4 |
|     9 | Daniel  | Clark | Bristol    | 1997-04-09 | M    | Analyst            |
    1 |
|    10 | Olivia  | Hall  | York       | 1995-08-14 | F    | Software Developer |
    2 |
+-------+---------+-------+------------+------------+------+--------------------+-
-------+
```

Department table:

```
MariaDB [project_schema]>create table Department (deptNo int primary key, deptName
varchar (30), mgrEmpNo int, foreign key (mgrEmpNo) references Employee(empNo));
Query OK, 0 rows affected (0.016 sec)


MariaDB [project_schema]> select * from Department;
+--------+-------------+----------+
| deptNo | deptName    | mgrEmpNo |
```

```
+--------+------------+---------+
|      1 | IT         |       1 |
|      2 | HR         |       2 |
|      3 | Finance    |       7 |
|      4 | Marketing  |       8 |
|      5 | Operations |       1 |
|      6 | Support    |       2 |
|      7 | Research   |       7 |
|      8 | Development|       8 |
|      9 | Security   |       1 |
|     10 | Admin      |       2 |
+--------+------------+---------+
```

Project table:

```
MariaDB [project_schema]>create table Project (projNo varchar(10) primary key,
projName varchar(30), deptNo int, foreign key (deptNo) references
Department(deptNo));
Query OK, 0 rows affected (0.018 sec)

MariaDB [project_schema]> select * from project;
+--------+-----------------------+--------+
| projNo | projName              | deptNo |
+--------+-----------------------+--------+
| 1      | Payroll System        |      1 |
| 10     | Office Automation     |     10 |
| 2      | Recruitment Portal    |      2 |
| 3      | Banking App           |      3 |
| 4      | Ad Campaign           |      4 |
| 5      | Logistics System      |      5 |
| 6      | Helpdesk Tool         |      6 |
| 7      | AI Research           |      7 |
| 8      | Mobile App            |      8 |
| 9      | Cyber Security Upgrade|      9 |
| AI01   | AI Project            |      3 |
| CLD1   | Cloud Migration       |      9 |
| DBMS   | Database System       |      2 |
| ERP1   | ERP System            |      8 |
| MOB1   | Mobile App            |      7 |
| NET1   | Network Upgrade       |      5 |
| RND1   | Research Project      |     10 |
| SCCS   | Course Control System |      1 |
| SEC1   | Security Upgrade      |      6 |
| WEB1   | Web Development       |      4 |
+--------+-----------------------+--------+
```

WorksOn table:

```
MariaDB [project_schema]> create table WorksOn ( empNo int,projNo varchar(10),
dateWorked date, hoursWorked int check( hoursWorked between 0 and 40), primary key
(empNo, projNo,dateWorked), foreign key (empNo) references Employee (empNo),
```

```
foreign key (projNo) references Project (projNo))

MariaDB [project_schema]> select * from WorksOn;

+-------+--------+------------+-------------+
| empNo | projNo | dateWorked | hoursWorked |
+-------+--------+------------+-------------+
|     1 | SCCS   | 2026-06-01 |           8 |
|     2 | DBMS   | 2026-06-02 |           6 |
|     3 | AI01   | 2026-06-03 |           7 |
|     4 | WEB1   | 2026-06-04 |           5 |
|     5 | NET1   | 2026-06-05 |           9 |
|     6 | SEC1   | 2026-06-06 |           4 |
|     7 | RND1   | 2026-06-07 |           8 |
|     8 | CLD1   | 2026-06-08 |           6 |
|     9 | ERP1   | 2026-06-09 |           7 |
|    10 | MOB1   | 2026-06-10 |           5 |
+-------+--------+------------+-------------+

MariaDB [project_schema]> show tables;
Output:
+--------------------------+
| Tables_in_project_schema |
+--------------------------+
| department               |
| employee                 |
| project                  |
| workson                  |
+--------------------------+
4 rows in set (0.016 sec)

7.25
MariaDB [project_schema]> create view projects_managed_by_females as select
p.projNo, p.projName from project p join department d on p.deptNo = d.deptNo join
employee e on d.mgrEmpNo = e.empNo where e.sex = 'F' order by p.projNo;
Query OK, 0 rows affected (0.014 sec)

MariaDB [project_schema]> select * from projects_managed_by_females;
+--------+--------------------+
| projNo | projName           |
+--------+--------------------+
| 10     | Office Automation  |
| 2      | Recruitment Portal |
| 4      | Ad Campaign        |
| 6      | Helpdesk Tool      |
| 8      | Mobile App         |
| DBMS   | Database System    |
| ERP1   | ERP System         |
| RND1   | Research Project   |
| SEC1   | Security Upgrade   |
```

```
| WEB1   | Web Development    |
+--------+--------------------+
```

7.26

```
create view view_2 as select e.empNo, e.fName,e.lName,w.hoursWorked,w.projNo from
Employee e join WorksOn w on e.empNo = w.empNo;
Query OK, 0 rows affected (0.011 sec)

MariaDB [project_schema]> select * from view_2;
+-------+---------+--------+-------------+--------+
| empNo | fName   | lName  | hoursWorked | projNo |
+-------+---------+--------+-------------+--------+
|     1 | John    | Smith  |           8 | SCCS   |
|     2 | Emma    | Brown  |           6 | DBMS   |
|     3 | Michael | Lee    |           7 | AI01   |
|     4 | Sarah   | White  |           5 | WEB1   |
|     5 | David   | Green  |           9 | NET1   |
|     6 | Anna    | Taylor |           4 | SEC1   |
|     7 | James   | Wilson |           8 | RND1   |
|     8 | Sophia  | Moore  |           6 | CLD1   |
|     9 | Daniel  | Clark  |           7 | ERP1   |
|    10 | Olivia  | Hall   |           5 | MOB1   |
+-------+---------+--------+-------------+--------+
```

7.27

```
CREATE VIEW EmpProject (empNo, projNo, totalHours) as select w.empNo,w.projNo,
SUM(w.hoursWorked) as totalHours from WorksOn w join Employee e ON e.empNo =
w.empNo join Project p onp.projNo = w.projNo GROUP by w.empNo, w.projNo;
```

a)

```
MariaDB [project_schema]> select * from EmpProject;
+-------+--------+------------+
| empNo | projNo | totalHours |
+-------+--------+------------+
|     1 | SCCS   |          8 |
|     2 | DBMS   |          6 |
|     3 | AI01   |          7 |
|     4 | WEB1   |          5 |
|     5 | NET1   |          9 |
|     6 | SEC1   |          4 |
|     7 | RND1   |          8 |
|     8 | CLD1   |          6 |
|     9 | ERP1   |          7 |
|    10 | MOB1   |          5 |
+-------+--------+------------+
10 rows in set (0.005 sec)
```

```
b)MariaDB [project_schema]> select projNo from EmpProject where projNo = 'SCCS';
```

```
+--------+
| projNo |
+--------+
| SCCS   |
+--------+
1 row in set (0.006 sec)

c)MariaDB [project_schema]> select count(projNo) from EmpProject where empNo='E1';
+---------------+
| count(projNo) |
+---------------+
|             0 |
+---------------+
1 row in set (0.005 sec)

d)MariaDB [project_schema]> select empNo , totalHours from EmpProject group by
empNo;
+-------+------------+
| empNo | totalHours |
+-------+------------+
|     1 |          8 |
|     2 |          6 |
|     3 |          7 |
|     4 |          5 |
|     5 |          9 |
|     6 |          4 |
|     7 |          8 |
|     8 |          6 |
|     9 |          7 |
|    10 |          5 |
+-------+------------+
10 rows in set (0.001 sec)
```

7.28

I would maintain this materialized view by storing the query results in a separate table and updating it via triggers on the Part table. An AFTER INSERT trigger would add a part number when partCost > 1000, an AFTER UPDATE trigger would insert or remove it depending on the updated cost, and an AFTER DELETE trigger would remove it if the part is deleted. You wouldn't need to access the base table during normal operations because, when things are updated, the triggers handle them. The base table will only need to be accessed at the beginning when the view is being populated and in rare cases.